

Experimental plan for the redo

Coupled magnetic–mechanical oscillator: one run of 8 separations $\times$ 5 trials

Planning document, version 3 — numbers are *predictions* from the first data set after the time-base correction of §0 ($f_1 = 6.49$ Hz, $C = 1.33 \times 10^{-7} \text{ m}^5 \text{ s}^{-2}$, $\tau \approx 21$ s); they will be replaced by the measured values

0. First: the time base — everything in run 1 was 8 times too slow

Your new FFT (clip set to 240 s^{-1} in Clip Settings) puts the main peak at 6.51 Hz, with the second one at ≈ 8.3 Hz; run 1 gave 0.811 Hz for the same spring. $6.51/0.811 = 8.03$. The explanation is not physics: a 240 fps slow-motion file is *stored* for playback at 30 fps, and Tracker reads that playback rate unless the frame rate is typed into Clip Settings. Run 1 was analysed with the file’s 30 fps, so every time was stretched by $240/30 = 8$ and every frequency divided by 8. The springs really oscillate at ≈ 6.5 Hz (6 s of your record contain about 40 swings — count them), and the pair beats about once per second at this d , which is what you saw: many beats in 6 s.

What this changes and what it does not:

- **Unchanged:** every ratio. f_2/f_1 , $2\gamma/k$, the exponent of the ln–ln graph (-3.75 in run 1), the static bend δ , x_{20} , the non-linear shifts in %, the equal-amplitude prediction, all Tracker advice. The run-1 data stay valid — multiply all their frequencies by 8.
- **Rescaled:** $f_1 = 0.811 \rightarrow 6.49$ Hz; $C = (f_2^2 - f_1^2)d^5$ becomes $64\times$ larger, $1.33 \times 10^{-7} \text{ m}^5 \text{ s}^{-2}$; $k/m = \omega_1^2$ becomes $64\times$ larger (1660 s^{-2}); the decay time $\tau \approx 170$ s becomes ≈ 21 s; every beat period is 8 times shorter. So the records of the previous version of this plan (60–400 s) were 8 times too long, and the “soft spring” and gravity discussion shrinks to a few-per-cent side issue (§2).
- **Check it once more** before the redo: in Clip Settings the frame rate must read 240 and the frame dt 4.17×10^{-3} s for every clip (import the first clip’s `.trk` into the others, §3). Say so explicitly in the essay’s method — it is exactly the kind of systematic check examiners reward.

1 What is measured and why

quantity	how	used for
f_1, f_2 at each d	FFT of $x_1(t)$ and $x_2(t)$ from Tracker; also FFT of x_1+x_2 and x_1-x_2	main graph $\ln(f_2^2 - f_1^2)$ vs $\ln d$ (gradient -5); $f_{\text{avg}}, f_{\text{beat}}$
d (centre to centre)	$d = g+t$: gap g between magnet faces (digital calliper / photo with scale) + magnet thickness t	independent variable
f_a, f_b	each spring plucked alone (other magnet removed)	identical-spring check, detuned model, gives k/m
k	static load–deflection graph (§6)	absolute test of C ; non-linear model
$M, \mu L$	balance	effective mass $m = M + \frac{33}{140}\mu L$, cross-check of $m = k/\omega_1^2$
L, b, h	ruler / calliper / micrometer	$k = Ebh^3/4L^3$ cross-check (with the small gravity term of §2); Rayleigh factor
magnet D, t, B_r or μ_m	calliper; data sheet or pull-force / Hall probe	$\mu_m \approx B_r V/\mu_0$, finite-size correction

Why $f_2^2 - f_1^2$. Subtracting the two mode results $\omega_1^2 = k/m$, $\omega_2^2 = (k + 2\gamma)/m$ removes k : $f_2^2 - f_1^2 = \gamma/2\pi^2 m = Cd^{-5}$. The *exponent* therefore needs only frequencies and d ; k , m and μ_m are needed only for the absolute value of C . Angular frequencies are not required anywhere ($\omega = 2\pi f$ cancels in every ratio), so the FFT frequencies can be used directly; convert to ω only if you want to quote $k/m = \omega_1^2$.

2 Apparatus and set-up

1. **Base and clamps.** Non-magnetic base (wood/aluminium), two clamps. Mark the base with a mm scale or fix a steel rule *behind the plane of the magnets* (out of the plane, so it does not attract them).
2. **Springs.** Two strips from the same stock (same b and h — check h with a micrometer, $k \propto h^3$), same free length L above the clamp jaws. Which of your lengths to use is decided by the *single-spring frequency* f_a , not by L itself — see “Which spring length” below; the strip you have been using ($f_a \approx 6.5$ Hz) is fine. Measure L from the top of the jaw to the *centre of the magnet*; write down L before every session because re-clamping changes it.
3. **Magnets.** Glue (or double-sided tape) each magnet flat on the *inner face* of the strip with its centre a few mm below the tip, like poles facing. It does *not* have to be exactly at the very tip — what matters is that (i) both are at the same height, (ii) the strip length is quoted to the magnet centre, (iii) the position never changes between runs. Put a small high-contrast dot (white marker / sticker) on the visible side of each magnet: this is what Tracker follows.
4. **Alignment.** Magnets coaxial: same height (± 1 mm), same depth from the camera, faces parallel. Check by eye from above and from the front.
5. **Camera.** Phone on a tripod, lens at the height of the magnets, ≥ 0.8 m away and *square* to the plane of motion (parallax otherwise turns a straight motion into a curved one). Zoom until the two magnets plus ≈ 30 mm on either side fill the frame width. Fix focus and exposure (tap and hold). Bright, even light; no flicker (LED/daylight, avoid old fluorescent tubes at 240 fps).
6. **Scale.** A ruler in the plane of the magnets, visible in every video (calibration stick in Tracker). Since d is measured with a calliper this scale only sets the mm/pixel factor for amplitudes.

Things that silently ruin a run: magnets glued at different heights (they stop being coaxial and F is no longer along x); a steel rule, steel clamp screw or the phone’s own magnet within a few cm of the magnets; the clamps loosening between trials (f_1 drifts — re-measure the “Infinity” frequency at the end of the session); camera not square to the motion.

Which spring length?

The magnets do not care about L ; they care about how stiff each strip is compared with the magnetic stiffness, i.e. about the *frequency of one spring alone with its magnet*, f_a . Every prediction in this plan depends on f_a only, so compare your strips by f_a , not by length: attach one magnet (or Blu-Tack of the same mass), clamp the strip, film 10 s, FFT in Tracker. Only the *free* length above the jaws counts, so a long strip can always be clamped shorter; what you cannot fix afterwards is a thickness difference, so both strips must come from the same stock. With the magnets of run 1 and the ladder of §3 (records of 45 s), this is what each f_a gives:

f_a / Hz	$L/L_{\text{run 1}}$	$f_2 - f_1$ at 24.5 mm	$f_2 - f_1$ at 42 mm	T_{beat} at 42 mm	largest usable d	$2\gamma/k$ at 24.5 mm	δ at 24.5 mm
4.5	1.28	1.44 Hz	112 mHz	9.0 s	46 mm	0.74	2.3 mm
5.0	1.19	1.33 Hz	101 mHz	9.9 s	45 mm	0.60	1.8 mm
5.5	1.12	1.23 Hz	92 mHz	10.9 s	44 mm	0.50	1.5 mm
6.5 (run 1)	1.00	1.07 Hz	78 mHz	12.9 s	42 mm	0.36	1.1 mm
8.0	0.87	0.89 Hz	63 mHz	15.8 s	41 mm	0.24	0.7 mm
10.0	0.75	0.73 Hz	51 mHz	19.7 s	39 mm	0.15	0.5 mm

(“largest usable d ”: where $3T_{\text{beat}}$ reaches the 45 s record; $L/L_{\text{run 1}}$ from $f \propto L^{-3/2}$, approximate because the strip’s own mass also changes with L .)

Recommendation: keep the strip you have ($f_a \approx 6.5$ Hz), **or go at most one step softer** ($f_a \approx 5\text{--}5.5$ Hz, $\approx 10\text{--}15\%$ longer). Softer strips give a larger splitting at every d (the magnetic stiffness is a larger fraction of k) and so a slightly wider usable range, but the gain from 6.5 to 5.5 Hz is only 15 %, while δ and x_{20} grow and the strip is easier to disturb. Below $f_a \approx 4.5$ Hz the closest point is no longer weakly coupled ($2\gamma/k \rightarrow 1$, $\delta > 2$ mm). Stiffer strips ($f_a \geq 8$ Hz) shrink the effect everywhere and lose the last ladder points. The strongest argument for keeping the present strip is continuity: run 1 (frequencies $\times 8$) then becomes a genuine pilot study in the essay, directly comparable with the redo.

Gravity, for completeness. A vertical strip with a tip magnet is an inverted pendulum: gravity reduces the stiffness, $k_{\text{eff}} = k - (1.2M + 0.375 \mu L) g/L$ (tip-load shape; setting $k_{\text{eff}} = 0$ gives the Euler buckling load to 1 %). At $f_a = 6.5$ Hz the correction is $(1.2\text{--}1.6) g/(L\omega_a^2) \approx 3\text{--}6\%$ of k for $L = 0.15\text{--}0.25$ m — worth one sentence when $k = Ebh^3/4L^3$ is compared with the measured k , and the reason for measuring k with the strip vertical and the magnet on (§6), but nothing to design around. (In the previous version of this plan, with the 8-times-too-low frequencies, the same formula made gravity cancel two thirds of k ; that was an artefact of the time base.) Since $f_a \propto L^{-3/2}$, a 1 % difference in L or 0.3 % in h detunes the pair by 1.5 % or 1 %: tune the pair by sliding one strip in its clamp (§4).

3 Separations, amplitude and record length

Which separations

Choose them roughly *evenly spaced in $\ln d$* (equal steps on the log–log graph), from the closest the clamps allow (you measured 24.5 mm) to the largest d at which the two peaks are still separated in a record of practical length. Eight values ($\approx 8\%$ steps), predictions for $f_1 = 6.49$ Hz:

d / mm (target)	24.5	26.5	28.5	31	33.5	36	39	42
predicted f_2 / Hz	7.56	7.23	7.01	6.84	6.73	6.66	6.60	6.57
predicted $f_2 - f_1$ / Hz	1.07	0.74	0.52	0.35	0.24	0.17	0.11	0.078
predicted T_{beat} / s	0.93	1.35	1.9	2.9	4.2	6.0	8.9	12.8
minimum ($3T_{\text{beat}}$) / s	2.8	4.0	5.7	8.6	12.6	18	27	39
record length / s	45	45	45	45	45	45	45	45
beats in the record	48	33	24	16	11	7.5	5	3.5
$\pm \frac{1}{2}$ -bin error on $f_2^2 - f_1^2$ (naive FFT)	1.5 %	2.1 %	3.0 %	4.5 %	6.6 %	9 %	14 %	20 %
static outward bend δ / mm	1.1	0.8	0.6	0.43	0.31	0.23	0.17	0.13

With the real time base one record length serves all points: 45 s (10 800 frames at 240 fps) is $\geq 3T_{\text{beat}}$ everywhere, and longer is pointless because the amplitude has decayed to $\approx 10\%$ by then ($\tau \approx 21$ s; the FFT peak width is then set by the damping, $\sqrt{3}/\pi\tau \approx 0.026$ Hz, not by $1/T$). A constant T is convenient (the same bin width, $1/T = 0.022$ Hz, for all d , so one uncertainty column) but *not* required: the rule is $T \geq 3T_{\text{beat}}$ at each point, and the uncertainty is computed from the T actually analysed. A 35 s record at 24.5 mm (37 beats, 0.029 Hz bins) is more than enough and does not need repeating; only the 39 and 42 mm points genuinely need $\geq 30\text{--}40$ s. Do not extend the ladder beyond ≈ 44 mm: from 45 mm on even a Hann-windowed 45 s record no longer separates the peaks (the single-peak points of run 1 at 45–60 mm were this limit, not a discovery).

The exact values do not matter (you cannot set d precisely anyway because the springs bend outwards by δ when the second magnet is mounted): set the clamps, *then measure d as it is*. Do not go below $d \approx 2D$ (D = magnet diameter; dipole approximation) or below $g \approx 8$ mm.

One trick to reach smaller d if you want it. The clamps limit the *clamp* spacing d_0 , not d . Mount the magnets on the *inner* faces (already planned) and, if the clamp jaws are wide, offset each strip towards the inner edge of its jaw. Alternatively clamp the two strips in one wide clamp with a spacer block between them: $d_0 = \text{spacer} + h$.

Ladder order. Do the eight separations in *random* order, not from small to large, and mark the clamp positions on the base so that a spoiled point can be repeated. Move only clamp 2 to change d ; clamp 1, the camera and the amplitude stop then stay where they are. One trial of each d first, then the remaining four — if something is wrong you find out after 8 videos, not 40.

Release amplitude

Pull spring 1 *outwards* (away from spring 2) by $A = 5$ mm at the magnet, against a stop or a mark (below), hold still for ≈ 2 s (spring 2 settles), release. Same A in every trial: A is a controlled variable because the non-linear force shifts f_2 by an amount that depends on the amplitude:

d / mm	24.5	26.5	28.5	31	33.5	36	39	42
f_2 shift, $A = 5$ mm	+1.2 %	+0.9 %	+0.7 %	+0.4 %	+0.3 %	+0.2 %	+0.1 %	+0.06 %
f_2 shift, $A = 10$ mm	+4.6 %	+3.7 %	+2.8 %	+1.8 %	+1.2 %	+0.8 %	+0.5 %	+0.3 %

(anti-phase mode with separation swing $\pm A$; simulated with the full r^{-4} force; percentages independent of the time base). $A = 5$ mm keeps this at or below the FFT uncertainty except at the two closest points, while still giving ≈ 100 pixels of motion if the frame width is ≈ 120 mm; the residual shift is exactly what the non-linear model in the evaluation predicts, so it becomes a test rather than an error. Why outwards: pulling apart lowers the repulsion during the hold, so spring 2 is not pushed off-axis and the magnets can never touch.

How to pull by 5 mm without Tracker. Tracker is never needed *during* the session: it is run once, afterwards, on the 40 clips, and if the tripod is not moved a single calibration stick and set of axes serve all clips (analyse the first clip, save the `.trk`, then File $\triangleright$ Import $\triangleright$ Tracker file to bring the calibration and axes into each new clip). Nor should you measure the gap between the two tips while pulling — spring 2 follows spring 1 by $x_{20} = A\gamma/(k + \gamma)$ (≈ 0.8 mm at 24.5 mm, ≈ 0.06 mm at 42 mm), so the tip-to-tip distance grows by more than A and by a different amount at each d . Reference spring 1 to the base instead:

- **Recommended: a stop set once per separation with a 5 mm gauge.** Fix a small post on the base outside spring 1, at magnet height (a bolt in a slotted bracket, a block on Blu-Tack, or a wooden dowel in a clip — anything non-magnetic that you can slide and fix). After the clamps are set and both springs are at rest, lock a calliper at 5.00 mm and use its inside jaws (or a 5 mm drill shank / stack of card measured with the calliper) as a gauge: hold it against the outer face of strip 1, push the post up to the gauge, fix the post, remove the gauge. Time: 20 s, once per d (the rest position of strip 1 shifts by up to 1 mm between separations because δ changes, so the post must be re-set each time d changes, but not between trials). Every trial is then: pull strip 1 against the post with two fingertips, hold 2 s, open the fingers. A is identical for the five trials to ± 0.1 mm with nothing to read or aim at. After release the strip can never reach the post again: its outward excursion is bounded by A (energy argument — the initial state is the one with all the energy in the springs) and damping plus the imperfect re-alignment of the two modes keep the closest return a few tenths of a millimetre short, arriving at zero velocity. Check the first clip at 24.5 mm anyway (closest return, at the first beat maximum) and glue a scrap of felt on the post face if you ever see a tap.
- **Alternative: two pencil marks.** A card fixed behind magnet 1 (in the plane of the tracked dot) with a mm scale or two marks 5 mm apart; note the rest reading once per d , pull until the dot is at rest -5 . Sighting error ≈ 0.5 mm (10 % of A) is harmless: A affects only the peak heights and the small non-linear shift ($\propto A^2$: ± 10 % in A moves f_2 at 24.5 mm by ± 0.25 %, less elsewhere), never f_1 .
- Pull *at the magnet height*, not half-way down the strip (so the strip shape is the static tip-load curve assumed by the theory), and pull with fingertips on the outer face so that the release is an opening of the fingers, never a push.

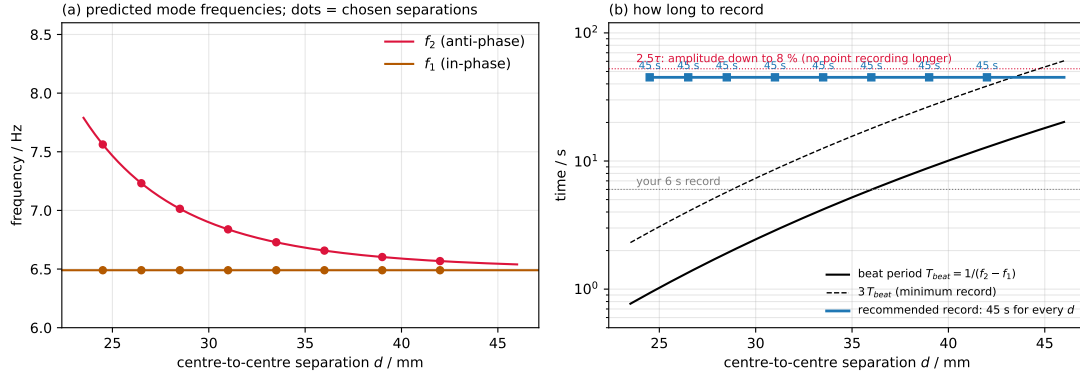


Figure 1: (a) Predicted mode frequencies at the chosen separations (real time base). (b) Beat period, the “three beats” minimum, the 45 s record, and the decay limit 2.5τ .

- The actual A of every trial comes free with the analysis — it is the first maximum of $x_1(t)$ after release — and is what you report in the data table; the stop or the marks only guide your hand.

Frame rate and record length

- **240 fps is right and now clearly needed:** at 6.5–7.6 Hz it gives 32–37 frames per oscillation; 30 fps would give only 4–5. In Tracker use **step size 2** (120 fps, 16–18 points per swing; the dot moves ≤ 2 mm between analysed frames, so the autotracker never loses it); step size 4 (8–9 points per swing) is still enough for the frequencies but marginal for the tracker at $A = 5$ mm. Type 240 into Clip Settings for every clip (§0).
- **Record length: 45 s for every d :** camera on, 3 s rest, pull, hold 2 s, release, then ≈ 45 s; stop at about one minute of video. The requirement is $T \geq 3T_{\text{beat}}$ (three swellings) *and* a resolution good enough for the smallest splitting: with $T = 45$ s the bin width is 0.022 Hz, versus $f_2 - f_1 = 0.078$ Hz at 42 mm (3.5 bins). Your 6 s record is *not* short because of the beats — it contains six of them — it is short because a 6 s record has 0.17 Hz bins: even at 24.5 mm the naive peak position is uncertain by ± 0.08 Hz, i.e. $\pm 11\%$ on $f_2^2 - f_1^2$, and at 42 mm the two peaks are one blob (Fig. 2). Beyond ≈ 50 s nothing is gained: the amplitude is down to 10% and the peak width is set by the damping.
- Quote T (the analysed window, starting ≈ 1 s after release) per point in the data table; it need not be the same for all points, but every point must satisfy $T \geq 3T_{\text{beat}}$.

4 The measurement session — step by step (8 separations $\times$ 5 trials)

Once per session

1. Photograph the set-up with the ruler; note date, room temperature, camera model, fps, distance camera–springs, which strip is “spring 1” (mark them with tape; they are never swapped).
2. Measure L (jaw to magnet centre), b , h (micrometer, 5 places along the strip), magnet D and t , mass M of each magnet with its glue/dot (weigh before gluing, or weigh identical spares), mass of a measured length of strip (μ).
3. **Tune the pair.** With the magnets far apart (> 70 mm), film each spring alone for 10 s and read f_a , f_b from the Tracker FFT (a stopwatch cannot resolve 1% at 6.5 Hz). If they differ by more than 1%, slide the faster (stiffer) strip a millimetre *up* in its clamp (or the slower one down; $f \propto L^{-3/2}$, so 1 mm on 150 mm is 1%) and repeat; then tighten fully and re-measure L .
4. **“Infinity” videos:** remove magnet 2 (or move spring 2 > 70 mm away, where the coupling shifts f by $< 0.1\%$), pull spring 1 by 5 mm, record 45 s; three trials. Same for spring 2 alone. Repeat at the *end* of the session to detect drift.

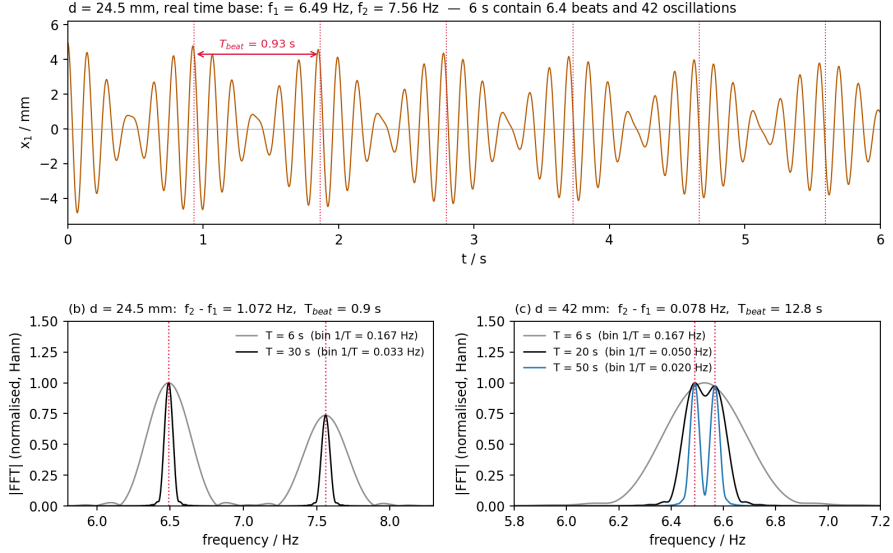


Figure 2: Why the record must be longer than 6 s even though 6 s already show six beats. Top: $x_1(t)$ at $d = 24.5$ mm (simulated with the run-1 constants). (b) Spectrum of a 6 s and of a 30 s record at 24.5 mm: both show two peaks, but the 6 s peaks are 0.17 Hz wide and their positions are uncertain by a good fraction of that. (c) At 42 mm a 6 s record gives one blob, 20 s a shoulder, 50 s two peaks.

Per separation (repeat for the 8 values, random order)

1. Move clamp 2 to the target spacing (clamp 1 stays); tighten; wait until both springs are at rest.
2. Measure the gap g between magnet faces at rest with a calliper (three readings, do not touch the magnets) or from a photo against the ruler; $d = g + t$.
3. Set the amplitude stop: calliper locked at 5.00 mm against the outer face of strip 1, slide the post up to it, fix, remove the calliper (§3).
4. Camera on, 3 s rest, pull strip 1 outwards against the stop, hold 2 s, release cleanly (open the fingers, do not push), record for the time in the table, stop. Do *not* touch the base while recording.
5. Repeat five times. Between trials wait until both springs are still (about 60 s of free decay, or damp them with a finger and wait 10 s; residual motion of spring 2 changes the initial condition).
6. Name files d24_t1.mp4 ...; keep a log line: file, g , d , T , anything unusual (hand touched, strip tapped the stop, wobble, draught).

More things to watch

- **Both springs at rest before every trial** — a residual 0.5 mm motion of spring 2 changes the mode amplitudes (not the frequencies, but it spoils the peak-ratio test). Damp them by touching both magnets briefly with a finger, then wait 10 s.
- **Pull slowly** (≈ 1 s), **hold** ≈ 2 s, the same every trial. A slow pull lets spring 2 follow quasi-statically to x_{20} ; a jerk would leave it ringing with up to 0.8 mm amplitude at 24.5 mm, and 2 s of hold damp that by only 10 % ($\tau \approx 21$ s). Pulled over 1 s, the residual ringing is ≈ 0.02 mm.
- **Storage:** 40 clips of one minute at 240 fps is ≈ 10 –40 GB depending on the phone; check the free space and that no clip is cut short.
- **Light flicker:** at 240 fps mains-powered lamps produce a 100 Hz brightness ripple that confuses autotracking. Use daylight or a battery lamp, and a matt background.
- **Temperature:** do all 40 videos in one session (about two hours) (Young’s modulus and remanence both drift $\sim 0.1\%$ /K); note the room temperature at the start and end.
- **Clamp creep:** after the session re-record the “Infinity” videos. If f_1 moved by more than 0.5 %, treat the drift as a systematic error (interpolate linearly in time between the two calibrations).

- **Keep the dots:** the tracked dot must not be repainted or moved between trials; if a magnet is knocked, re-measure g .
- **Tripod and camera:** don't touch the tripod during recording; use the volume button/remote or a 3 s timer to start; lock focus and exposure.

Release with a burnt thread instead of fingers?

Not needed here. The thread-and-flame release is for *vertical* pendulums where a hand disturbs the plane; for a leaf spring pulled sideways by 5 mm, letting go with two fingertips gives a clean start, and the analysis window begins 1 s later anyway. A thread would also have to be tied at the magnet, exactly where you track. If you prefer a mechanical release, a small wooden wedge pulled away horizontally works.

5 Instead of a second run: the data-processing comparison

Repeating the whole grid is not needed if the first run already follows the rules above; the “method development” can be done entirely in the analysis, on the same tracks, and it is worth more marks than a second identical run because it isolates the effect of each analysis choice. For every video compute f_1, f_2 with:

1. **Rectangular window, full record** — the naive FFT; peak positions from the highest bin. Expected uncertainty $\pm \frac{1}{2} \cdot 1/T$.
2. **Rectangular window, first 6 s only** — reproduces the situation of the first analyses (peaks 0.17 Hz wide, merged beyond ≈ 33 mm); shows why T matters.
3. **Hann window + zero padding** ($N_{\text{FFT}} \geq 2^{16}$ for 10 800 samples) — suppresses leakage between the two peaks; peak from the interpolated maximum; uncertainty $\approx 0.1/T \approx 2$ mHz for a clean peak.
4. **Normal coordinates** $x_A + x_B$ (only f_1) and $x_A - x_B$ (only f_2) — no resolution problem; requires identical springs, so its disagreement with method 3 is itself a measurement of the detuning.
5. **Two-tone least-squares fit** $x = [a \cos(2\pi f_1 t + \phi_1) + b \cos(2\pi f_2 t + \phi_2)] e^{-t/\tau}$ — model-based, sharpest, and gives a/b and τ directly; with $\tau \approx 21$ s the decay factor is not optional here.

Present one table per method (8 rows of mean $\pm$ SD over 5 trials), the resulting ln–ln gradient for each, and the comparison with the theoretical resolution $1/T$; then adopt method 3 (or 4) for the main result and explain why. This replaces “run 2” in the write-up, and the amplitude effect (§3) becomes the non-linear model test in the evaluation.

6 Static stiffness k (needed once; do it after the videos)

With $\omega_a^2 = (2\pi \times 6.49)^2 = 1660 \text{ s}^{-2}$ and a moving mass of 3–10 g, $k = m\omega_a^2 \approx 5\text{--}17 \text{ N m}^{-1}$: a 2 g load deflects the tip by 1–4 mm, so ordinary small masses work. Measure the stiffness that enters the theory — the strip *vertical*, its magnet on, the other magnet removed — so that the small gravity term (§2) is included automatically:

1. **Horizontal load over a pulley (standard).** Thread tied round the strip at the magnet centre, over a low-friction pulley, masses 1, 2, $\dots$, 10 g (weighed, ± 0.01 g) hung vertically; the thread must be horizontal. Tip deflection x from a photo against the ruler (or Tracker on the photo, ± 0.1 mm), loading and unloading to expose friction and hysteresis. $k = \text{gradient of } F = m_w g \text{ against } x$; keep $x \leq 10$ mm (linear region, and the amplitude range used in the videos).
2. **Tilt method (no pulley, friction-free).** Raise one edge of the base on shims of known thickness so that gravity has a component along x ; $\theta = \arctan(\text{rise}/\text{base width})$. With $k \approx 10 \text{ N m}^{-1}$ the tip moves only ≈ 0.1 mm per degree, so this needs θ up to $10\text{--}15^\circ$ and is less convenient here

than the pulley — use it only if no pulley is available. Generalised force for the tip-load shape:
 $F = g \sin \theta (M + 0.375 \mu L)$.

3. **Dynamic check.** Add known masses $\Delta m = 1, 2, 3$ g (weighed Blu-Tack) at the magnet and measure f_a each time. Plotting $1/(2\pi f_a)^2$ against Δm gives $1/k$ as the gradient and m/k as the intercept — a second, independent measurement of both k and m_{eff} (the gravity term changes by $1.2g\Delta m/L \approx 0.1\text{--}0.2 \text{ N m}^{-1}$, i.e. 1–2%: mention, do not fuss).

Then compare $m_{\text{eff}} = k/(2\pi f_a)^2$ with $M + \frac{33}{140}\mu L$, and k with $Ebh^3/4L^3$ (expect the measured value a few per cent lower: gravity, and a clamp is never perfectly rigid).

Moment of inertia is not needed: the magnet is treated as a point mass moving with the tip; its rotation contributes $\frac{1}{2}I\dot{\theta}^2$ with $\theta \approx 1.5x/L$, i.e. a fraction $\sim (I/M)(1.5/L)^2 \sim (D/L)^2/10 \lesssim 10^{-3}$ of the kinetic energy. Mention it as a negligible correction, do not measure it.

7 Data processing checklist

1. Tracker: Clip Settings with frame rate **240** typed in (check $\text{dt} = 4.17 \times 10^{-3} \text{ s}$ in every clip), start frame ≈ 1 s after release, end frame 45 s later, step size 2; calibration stick; axes with origin on the dot of magnet 1 in a rest frame, x horizontal; point mass A on magnet 1, point mass B on magnet 2 (autotracker); export **t**, **x_A**, **x_B** as text.
2. Python: subtract the mean of each track, then the five methods of §5; beat period from the envelope; peak-height ratios r_1, r_2 (identical-spring test); actual A from the first maximum; τ from the envelope.
3. **Equilibrium of mass B.** x_B oscillates about $\approx d$, not about 0, because the origin sits on magnet 1 — leave it. Do not move the axes to fix it (that would give the two masses different origins) and do not shift x_B in Tracker. In Python subtract from each track its own mean over the analysed window (over ≥ 3 whole beats the mean of a decaying two-tone signal is its rest position to $\ll 0.1$ mm), or subtract the value averaged over the 3 rest seconds before the pull; both give x_1, x_2 measured from equilibrium as in the theory. Tracker’s own FFT ignores the offset anyway (it appears only in the zero-frequency bin, which is why your spectrum is clean).
4. Per d : mean and standard deviation of the 5 trials (random error), compared with the FFT uncertainty (systematic floor); take the larger.
5. Graphs: f_1, f_2 vs d ; $\ln(f_2^2 - f_1^2)$ vs $\ln d$ with weighted fit (gradient vs -5); f_{avg} and f_{beat} vs d ; residuals; then the non-linear and finite-size models in the evaluation.

Summary card (print and keep at the bench)

fps	240 recorded and typed into Tracker’s Clip Settings ($\text{dt} = 4.17 \text{ ms}$); step size 2
spring	the present strip ($f_a \approx 6.5 \text{ Hz}$) or one $\leq 15\%$ longer; tune $f_a = f_b$ to $< 1\%$ (Tracker FFT of each spring alone) by sliding one strip in its clamp
release	spring 1 pulled outwards against the stop (set once per d with the calliper locked at 5.00 mm), hold 2 s, open the fingertips
d	set clamps, then <i>measure</i> g with calliper, $d = g + t$; 3 readings
record	3 s rest, slow pull (1 s), hold 2 s, release, then 45 s for $d \approx 24.5, 26.5, 28.5, 31, 33.5, 36, 39, 42 \text{ mm}$ (rule: $T \geq 3T_{\text{beat}}$ — 3.5 beats at 42 mm, 48 at 24.5 mm; the 35 s already recorded at 24.5 mm is fine; quote T per point)
trials	5 per d , random order of d ; wait until still between trials
also	“Infinity” (each spring alone, $> 70 \text{ mm}$ apart) $3 \times 45 \text{ s}$ at start and end; photo of set-up; L, b, h, M, μ, D, t ; k by load–deflection (strip vertical, magnet on)
never	steel/phone near the magnets; touching the base or tripod during a recording; changing L , the dots or the magnet positions during the session